PROBLEM CL16

Plane Jet Split by a Non-Symmetric Wedge: Force, Moment and Heat

Original statement · CL16.pdf

Free jets | Bernoulli | Moment of a jet | Heat transfer

OFFICIAL PROBLEM STATEMENT

A plane liquid jet of constant density ρ, thickness h and uniform velocity v, impinges on a non-symmetric wedge of geometry and dimensions shown in the figure of CL16.pdf. The wedge splits the jet into two layers of unequal flux, such that a portion α of the volume flux is deflected an angle θ, and the rest of the fluid is deflected an angle 2θ. The flow is steady; viscous and gravitational forces are negligible; the known data are $(\rho, c, v, h, \alpha, \theta, L, p_a, T_0, T_1, T_2)$.

- Noting that the conditions for the use of Bernoulli's theorem are met, determine the (uniform) velocity of the fluid on each deflected layer.
- Determine the thickness h_1 and h_2 of each layer.
- Determine the force $\bar{F}$ (per unit length in the z-direction) that the liquid jet and the ambient air exert on the wedge.
- Determine the moment $\bar{M}_0$ due to the force $\bar{F}$ about the corner $\bar{x}_0$ of the wedge. Find the value of the fraction α that would produce a null moment, $\bar{M}_0 = 0$.
- Knowing the uniform temperature T_0 of the incoming jet, the uniform temperatures T_1 and T_2 of the two liquid layers when they lose contact with the wedge, and the constant specific heat c of the liquid, determine the heat q transferred per unit surface from the liquid to the wedge. Assume that the heat flux is uniform on the wedge surface and that there is no heat transfer by conduction between the liquid and the ambient air.

Source: CL16.pdf (pages 1-2, exam of Jan 14, 2013, with handwritten results). Notes.pdf: Bernoulli's equation Eq. 4.17, integral mass balance Eq. 3.11, integral momentum balance Eq. 3.36, integral angular momentum balance Eq. 3.38, integral total energy balance Eq. 3.46.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Steady, planar (per unit length in z), incompressible, frictionless; neglect $\bar{g}$; free jets at p_a (straight parallel streamlines in the layers away from the corner, so $p = p_a$ inside them).

- Geometry (figure): the incoming jet is centred on the x axis, which passes through the corner $\bar{x}_0$; layer 1 leaves along $\bar{t}_1 = \cos\theta\,\bar{e}_x + \sin\theta\,\bar{e}_y$ on the upper face, layer 2 along $\bar{t}_2 = \cos 2\theta\,\bar{e}_x - \sin 2\theta\,\bar{e}_y$ on the lower face. The faces have horizontal length L, i.e. lengths $L/\cos\theta$ and $L/\cos 2\theta$.
- Heat: uniform flux q from the liquid to the wedge on both faces; no conduction to the air; $e = cT$.
- Data: $[\rho,\ c,\ v,\ h,\ \alpha,\ \theta,\ L,\ p_a,\ T_0,\ T_1,\ T_2]$. Unknowns: $[v_1,\ v_2,\ h_1,\ h_2,\ \bar{F},\ \bar{M}_0,\ q]$.

Symbol	Meaning	SI unit
v, h	jet velocity and thickness	m/s, m
α, θ	flux fraction and deflection angle	1, rad
h_1, h_2	layer thicknesses	m
$\bar{F}$, $\bar{M}_0$	force and moment per unit length	N/m, N·m/m
q	heat flux to the wedge	W/m^2

PHASE 2

Frames, Control Volume and Surfaces

- Frame fixed to the wedge; origin at the corner $\bar{x}_0$; $\bar{e}_z$ towards the reader.
- Surfaces: Σ_0 (incoming jet section, $\bar{n} = -\bar{e}_x$), Σ_1, Σ_2 (layer sections normal to $\bar{t}_1, \bar{t}_2$, $\bar{n} = \bar{t}_1, \bar{t}_2$), Σ_a (free surfaces, $p = p_a$) and Σ_w (wetted faces). On layer k, $\bar{x} - \bar{x}_0 = s\,\bar{t}_k + \eta\,\bar{m}_k$, with $\bar{m}_1 = -\sin\theta\,\bar{e}_x + \cos\theta\,\bar{e}_y$, $\bar{m}_2 = -\sin 2\theta\,\bar{e}_x - \cos 2\theta\,\bar{e}_y$ the unit normals pointing away from the wedge and $0 \le \eta \le h_k$.

$$\boxed{\bar{F} = \int_{\Sigma_w} \left[(p - p_a)\,\bar{n} - \bar{\bar{\tau}}' \cdot \bar{n}\right] d\sigma}$$

$$\boxed{\bar{M}_0 = \int_{\Sigma_w} (\bar{x} - \bar{x}_0) \wedge \left[(p - p_a)\,\bar{n} - \bar{\bar{\tau}}' \cdot \bar{n}\right] d\sigma}$$

$$\boxed{q = -\frac{1}{L/\cos\theta + L/\cos 2\theta} \int_{\Sigma_w} \bar{q} \cdot \bar{n}_{\text{in}}\, d\sigma}$$

PHASE 3

Bernoulli and Balances, Surface by Surface

1. Velocities

Bernoulli (Notes Eq. 4.17, $U = 0$) along a free streamline from the incoming jet to each layer, with $p = p_a$ at both ends: $p_a + \frac{1}{2}\rho v^2 = p_a + \frac{1}{2}\rho v_k^2$, so

$$\boxed{v_1 = v_2 = v} \tag{1}$$

> 2. Mass (Notes Eq. 3.11): thicknesses
>
> $$\underbrace{\frac{d}{dt}\cancel{\int_{V_c} \rho\, dV}}_{0\ \text{(steady)}} + \int_{\Sigma_c} \rho\,(\bar{v} - \underbrace{\cancel{\bar{v}_c}}_{0}) \cdot \bar{n}\, d\sigma = 0$$
>
> Each surface integral, surface by surface (the vertical line drawn under each integral in class):
>
> | **Surface** | $\int \rho\, \bar{v} \cdot \bar{n}\, d\sigma$ |
> |---|---|
> | Σ_0 | $-\rho v h = -\rho Q$ |
> | Σ_1 | $+\rho v h_1 = \rho \alpha Q$ |
> | Σ_2 | $+\rho v h_2 = \rho(1-\alpha) Q$ |
> | $\Sigma_a,\ \Sigma_w$ | 0 |
>
> $$\boxed{h_1 = \alpha h, \qquad h_2 = (1-\alpha) h} \tag{2}$$

3. Linear momentum (Notes Eq. 3.36)

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c}\rho\bar{v}\,dV}}_{0\ (\text{steady})}+\int_{\Sigma_c}\rho\bar{v}\,[(\bar{v}-\underbrace{\bar{v}_c}_{0})\cdot\bar{n}]\,d\sigma=-\int_{\Sigma_c}(p-p_a)\,\bar{n}\,d\sigma+\int_{\Sigma_c}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma+\underbrace{\cancel{\int_{V_c}\rho\bar{g}\,dV}}_{\text{neglect }\bar{g}}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\bar{v}\,(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int(p-p_a)\,\bar{n}\,d\sigma$	$\int\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$
Σ_0	$-\rho v^2 h\,\bar{e}_x$	0	0
Σ_1	$\rho v^2\alpha h\,(\cos\theta\,\bar{e}_x+\sin\theta\,\bar{e}_y)$	0	0
Σ_2	$\rho v^2(1-\alpha)h\,(\cos 2\theta\,\bar{e}_x-\sin 2\theta\,\bar{e}_y)$	0	0
Σ_a	0	$0\ (p=p_a)$	0
Σ_w	0	$-\int_{\Sigma_w}(p-p_a)\,\bar{n}\,d\sigma$	$\int_{\Sigma_w}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$

Boxed Σ_w terms: $-\bar{F}$.

$$\boxed{\bar{F}=\rho v^2 h\,\{[1-\alpha\cos\theta-(1-\alpha)\cos 2\theta]\,\bar{e}_x-[\alpha\sin\theta-(1-\alpha)\sin 2\theta]\,\bar{e}_y\}}\quad(3)$$

4. Angular momentum about $\bar{x_0}$ (Notes Eq. 3.38)

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c}\rho\,(\bar{x}-\bar{x}_0)\wedge\bar{v}\,dV}}_{0}+\int_{\Sigma_c}\rho\,[(\bar{x}-\bar{x}_0)\wedge\bar{v}](\bar{v}\cdot\bar{n})\,d\sigma=-\int_{\Sigma_c}(\bar{x}-\bar{x}_0)\wedge(p-p_a)\bar{n}\,d\sigma+\int_{\Sigma_c}(\bar{x}-\bar{x}_0)\wedge(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int\rho\,[(\bar{x}-\bar{x}_0)\wedge\bar{v}](\bar{v}\cdot\bar{n})\,d\sigma$	pressure and viscous moments
Σ_0	$\int_{-h/2}^{h/2}\rho v\,(-yv)(-1)\,dy\,\bar{e}_z=0$ (odd in y)	0
Σ_1	$\int_0^{h_1}\rho v^2\,\eta\,(\bar{m}_1\wedge\bar{t}_1)\,d\eta=-\rho v^2\frac{h_1^2}{2}\,\bar{e}_z$	0
Σ_2	$\int_0^{h_2}\rho v^2\,\eta\,(\bar{m}_2\wedge\bar{t}_2)\,d\eta=+\rho v^2\frac{h_2^2}{2}\,\bar{e}_z$	0
Σ_w	0	$-\bar{M}_0$

Here $\bar{m}_1 \wedge \bar{t}_1 = (-\sin^2\theta - \cos^2\theta)\,\bar{e}_z = -\bar{e}_z$ and $\bar{m}_2 \wedge \bar{t}_2 = (\sin^2 2\theta + \cos^2 2\theta)\,\bar{e}_z = +\bar{e}_z$; the part $s\,\bar{t}_k \wedge \bar{t}_k$ vanishes. The antiderivative of η is $\eta^2/2$; Barrow's rule between 0 and h_k gives $h_k^2/2$. With (2): $-\frac{\rho v^2 h^2}{2}\left[\alpha^2 - (1-\alpha)^2\right] = -\frac{\rho v^2 h^2}{2}(2\alpha - 1)$.

$$-\frac{\rho v^2 h^2}{2}(2\alpha-1)\,\bar{e}_z = -\bar{M}_0 \implies \boxed{\bar{M}_0 = \rho v^2\,\frac{h^2}{2}\,(2\alpha-1)\,\bar{e}_z, \qquad \bar{M}_0 = 0 \iff \alpha = \frac{1}{2}} \quad (4)$$

5. Total energy (Notes Eq. 3.46)

$$\underbrace{\frac{d}{dt}\int_{V_c} \rho\left(e + \tfrac{v^2}{2}\right) dV}_{0\ \text{(steady)}} + \int_{\Sigma_c} \rho\left(e + \frac{v^2}{2}\right)\left[(\bar{v} - \underbrace{\bar{v}_c}_{0})\cdot\bar{n}\right] d\sigma = -\int_{\Sigma_c} p\,\bar{v}\cdot\bar{n}\,d\sigma + \int_{\Sigma_c} \bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma + \underbrace{\int_{V_c} \rho\bar{g}\cdot\bar{v}\,dV}_{\text{neglect } \bar{g}} - \int_{\Sigma_c} \bar{q}\cdot\bar{n}\,d\sigma + \underbrace{\int_{V_c} Q_H\,dV}_{0}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\left(e + \frac{v^2}{2}\right)(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int p\,\bar{v}\cdot\bar{n}\,d\sigma$	$\int \bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$	$-\int \bar{q}\cdot\bar{n}\,d\sigma$
Σ_0	$-\rho v h\left(cT_0 + \frac{v^2}{2}\right)$	$+p_a v h$	0	0
Σ_1	$+\rho v\alpha h\left(cT_1 + \frac{v^2}{2}\right)$	$-p_a v\alpha h$	0	0
Σ_2	$+\rho v(1-\alpha) h\left(cT_2 + \frac{v^2}{2}\right)$	$-p_a v(1-\alpha)h$	0	0
Σ_a	0	0	0	0 (no conduction to the air)
Σ_w	0	0 ($\bar{v} = 0$)	0	$-q\left(\frac{L}{\cos\theta} + \frac{L}{\cos 2\theta}\right)$

Boxed Σ_w term: the heat leaves the liquid towards the wedge. The kinetic-energy and pressure-work terms cancel because $\alpha + (1-\alpha) = 1$.

$$\rho v h c\left[\alpha T_1 + (1-\alpha)T_2 - T_0\right] = -q\left(\frac{L}{\cos\theta} + \frac{L}{\cos 2\theta}\right) \implies \boxed{q = \frac{\rho c v h}{\frac{L}{\cos\theta} + \frac{L}{\cos 2\theta}}\left[T_0 - \alpha T_1 - (1-\alpha)T_2\right]} \quad (5)$$

PHASE 4

Interpretation, Limits and Dimensional Check

- Symmetric-split check: with $\alpha = 1/2$ the two layers carry equal flux and their moments about the corner cancel, (4). The force (3) is still not symmetric because the deflections θ and 2θ differ.
- Limit $\theta \to 0$: no deflection, $\bar{F} \to 0$ in (3) (both brackets vanish).
- (5): $q > 0$ (liquid heats the wedge) when the liquid leaves colder than it arrived on average.
- Dimensions (SI, per unit length): $[\rho v^2 h] = \mathrm{N/m}$; $[\rho v^2 h^2] = \mathrm{N \cdot m/m}$; $[\rho c v h\, \Delta T/L] = \mathrm{W/m^2}$.
- Numerical cross-check of (4) (hypothetical data): $\rho = 1000$ kg/m^3, $v = 10$ m/s, $h = 0.02$ m, $\alpha = 0.3$, $\theta = 25^\circ$: midpoint quadrature of the two layer integrals gives an angular-momentum flux of 8.000 N·m/m, equal to $-\rho v^2 h^2 (2\alpha - 1)/2 = 8$, i.e. $M_{0,z} = -8$ N·m/m.
- Agreement with the official solution: CL16.pdf obtains (1) to (5).